PROJECT INTRODUCTION · SEE THE FUTURE 2026 · CIRCULAR ECONOMY

Handoff

A verified campus reuse board where students give, sell, lend or swap what they already own — and every confirmed handoff is measured.

handoff-bay-two.vercel.app · github.com/mn3265-commits/handoff

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THE PROBLEM

A campus throws out most of a year in one week, and buys it back in September.

At move-out, working lamps, fridges, rugs, textbooks and coats go into the same dumpsters — not because students want to discard them, but because listing an item takes longer than carrying it downstairs, and because the alternative is meeting a stranger from a public marketplace off campus. Eight weeks later the next intake buys the identical objects new, often financed on money they do not have.

12.1M t
US furniture and furnishings generated in 2018 — 80.1% landfilled (EPA)

17M t
US textiles generated the same year; 11.3M t landfilled (EPA)

1 week
of May carries most of a residential campus's annual discard

Existing responses are one-week, staffed, one-way donation drives, bound by trucks, storage and volunteer hours. Nothing runs on the ordinary Tuesday when the decision to bin the chair is actually made. **The waste is downstream of a user-experience problem, and that is the part nobody has built for.**

THE SOLUTION

Four ways to keep an object in use, and one tap that counts it.

Twenty seconds to post

One photo, then one sentence written like a text message. The app reads price, category, condition and meetup spot out of the words — no form — and any of it is corrected in one tap.

Give, sell, lend or swap

Ownership is one way to have a thing. A drill borrowed by six people is five drills never manufactured; a swap keeps two objects in use instead of one.

Only your campus sees it

One verified university email per account, enforced in the database. Every school is an isolated board — which removes the stranger problem that stops students listing.

The community is the logistics

A fridge is not a one-person object. Listings can ask for two people or a trolley, and students on the same campus take that work for a fee paid directly between them — no van dispatched, so the transport term stays zero.

Both sides confirm

The exchange happens at a library entrance or a dining hall door. Both people tap "handed off": the object is counted, and each public handoff count rises by one.

WHY IT IS DIFFERENT

Reuse measured at the object, not at the truck.

- **An auditable circularity metric.** Drives report a weight after the fact, in aggregate. Handoff records one confirmed event per object, with category and timestamp, agreed by both parties — bottom-up, and defensible line by line.
- **The measurement is the reputation.** The tap that counts the reuse is the tap that raises both people's public handoff count, so the incentive to record impact is intrinsic rather than bolted on for reporting.
- **Heavy things move without a vehicle.** The wall every reuse venture hits is the fridge nobody can carry. A campus already contains the answer: a student with an hour and a trolley, paid directly, with a public carry count of their own.
- **No transport term.** A collection round burns fuel per item — roughly 2.5 kg CO_2e for a 10 km round trip, at the EPA's passenger-car factor, which flatters a van. Two students crossing a campus they were crossing anyway emit nothing extra.
- **An honest model, published in the open.** Measured, estimated and conservative numbers are kept apart; factors take the low end; displacement is held at 0.5; and a borrowed object earns no carbon credit at all, because it comes back.
- **Any university, instantly.** A school with no board gets one the moment its first student signs in — the campus is created, named and marked from a registry of 7,328 academic domains. No waiting list, no launch.

THE IMPACT MODEL · V1.1

Three levels, never blurred.

LEVEL	NUMBER	STATUS
1	Confirmed handoffs — both people tapped, one object, one timestamp	**Measured**
2	Mass kept in use = count × typical mass for the category	Estimated
3	Production avoided = mass × low-end cradle-to-gate factor × displacement 0.5	Estimated, conservative

Forecasts use an assumed move-out category mix weighted toward small things rather than an unweighted average, which lowers our own estimate by roughly a third. On that basis a single participating floor — 120 students, three items each — is about **360 objects, 1.2 tonnes kept out of landfill and 2.6 tonnes CO_2e of avoided production**. The full factor table, the mix and the sources are printed on the public site, and the model lives in one file in an open repository.

CURRENT STAGE

Working product, pre-pilot.

Deployed and running on a live backend, not a mockup: Google (university) sign-in, campus-isolated Postgres with row-level security, realtime chat, photo storage, the day-7 freshness check, a community-rules gate, a pre-post safety check, and the both-sides handoff confirmation. The board is also an installable PWA and a full desktop web app. Two verified accounts exist; the board has not yet been opened to residents. The pilot is scoped for the coming term, ahead of the May 2027 move-out peak.

HOW IT PAYS FOR ITSELF

Charged before a deal exists, never on the object.

Campus licence

The main line. Housing and sustainability offices already pay to haul this material away and already must report circularity; a licensed campus gets admin tools, the move-out programme and an audited object-level number.

Verified-audience placements

Movers, storage and student services reaching one campus at one fortnight of the year. Paid up front, clearly marked, never mixed into ranking.

Free permanently

Posting, claiming, giving, lending, swapping. No listing fee and no cut of a handoff — a fee on a give-away would be a tax on generosity.

TEAM

Two people, in the open.

Agung Nugroho — product and build. Designs the board, writes the code, owns the impact model, runs the pilot. Graduate student and Graduate Assistant at Columbia SPS; professional background in customer-lifecycle and CRM marketing at telecommunications scale, where the daily problem is the same one this product has: getting a very large number of ordinary people to complete one specific action, and measuring honestly whether they did.

Tessa Wong — Columbia University. [role and background]

Campus reuse does not fail on technology. It fails on adoption and on trust, which are behaviour and product problems before they are engineering ones.

SCALE

Dense before wide.

Adding a school costs nothing and requires no code: a university email opens a board, and row-level security keeps every campus isolated. What does not copy is density, so our attention stays on one campus — building by building — until the board is useful on an ordinary Tuesday and not only in the last week of May. Distribution then runs through institutions rather than advertising: housing, sustainability offices and student groups already own the move-out problem, and the impact model is the pitch to them.
